

NAME: SAVITHA R

REG.NO: 192421368

15: Implement Perceptron based IRIS classification

Program:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix

# Load Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Perceptron classifier
model = Perceptron(max_iter=1000, eta0=0.1)

# Train the model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)
```

```
# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Calculate confusion matrix
cm = confusion_matrix(y_test, y_pred)

# Display results
print("Actual Values:")
print(y_test)

print("\nPredicted Values:")
print(y_pred)

print("\nAccuracy:")
print(accuracy)

print("\nConfusion Matrix:")
print(cm)

# Predict a new flower
new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_flower)

print("\nNew Flower:")
print(new_flower)

print("Predicted Class:")
print(prediction[0])
```

```
print("Flower Name:")  
print(iris.target_names[prediction[0]])
```

output:

```
> Enter help below or click help above for more information.  
===== RESTART: E:/Machine Learning/exp 18.py =====  
Actual Values:  
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0 0 0 1 0 0 2 1  
 0 0 0 2 1 1 0 0]  
  
Predicted Values:  
[2 0 2 2 1 0 1 2 2 1 2 0 0 0 0 2 2 1 2 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 2  
 0 0 0 2 2 2 0 0]  
  
Accuracy:  
0.8222222222222222  
  
Confusion Matrix:  
[[19  0  0]  
 [ 0  5  8]  
 [ 0  0 13]]  
  
New Flower:  
[[5.1, 3.5, 1.4, 0.2]]  
Predicted Class:  
0  
Flower Name:  
setosa
```